

# Multivariate OLS

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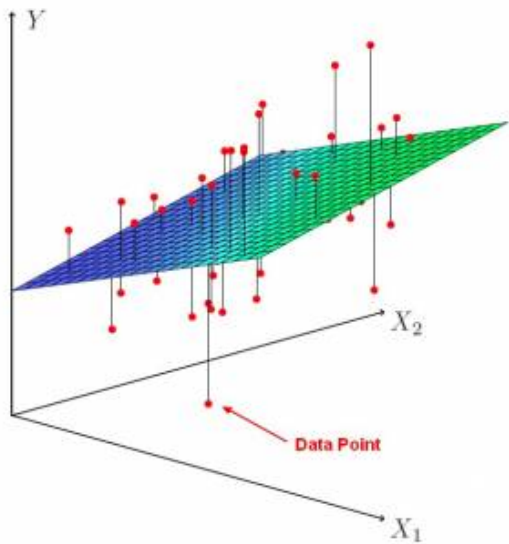
Session 4

# Multivariate Regression

- What if the true data generating process is multivariate, rather than bivariate?
- $y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \epsilon$
- If we run a bivariate regression, we have incorrectly specified the data generating process by omitting a causal factor from our model.
- What are the consequences of adding lots of regressors?
  - ▶ Surely there must be consequences to throwing variables in that aren't related to  $Y$ ?
  - ▶ Surely there must be consequences to omitting variables that are?

# Multivariate Regression

- Multivariate regression allows us to see the effects of two or more variables on our dependent variable.
- What is the effect of  $x$  on  $y$ , controlling for the effects of  $z$ ?
- We specify a model — a data-generating process — with more than one independent variable.
- We fit a plane or hyperplane.



# Why control for another variable

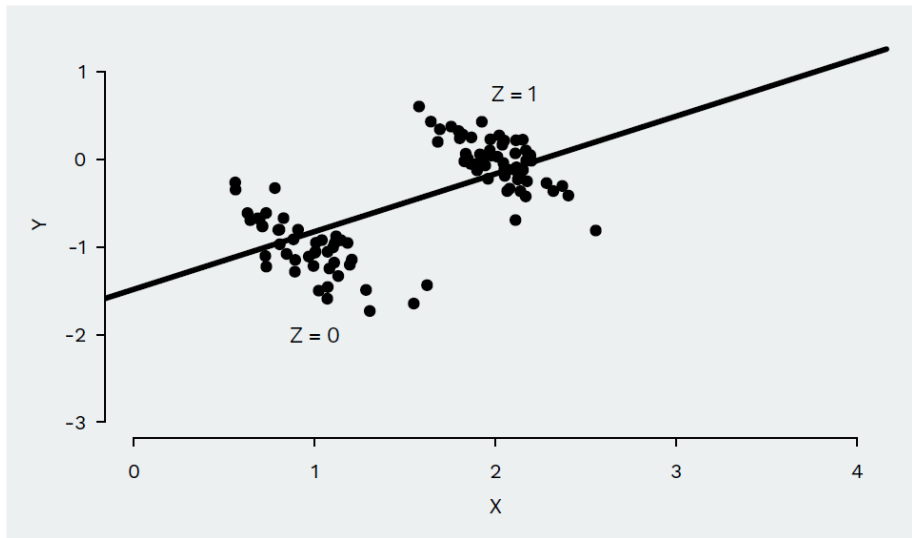
Predictive:

- We can usually make better predictions about the dependent variable with more information on independent variables.

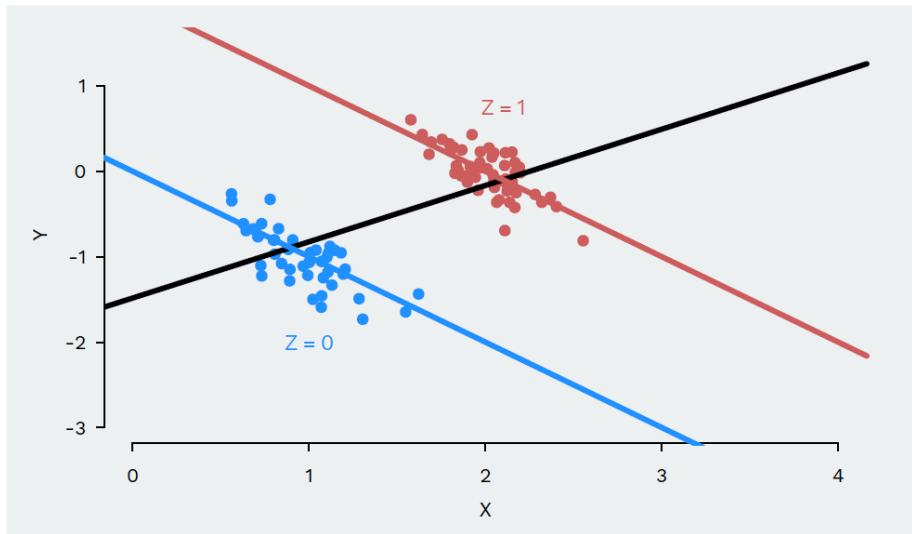
Causal:

- Block potential confounding, which is when  $X$  doesn't cause  $Y$ , but only appears to because a third variable  $Z$  causally affects both of them.

# Simpson's paradox



# Simpson's paradox



# The linear model

- The general linear model extends simple linear regression (SLR) by adding terms linearly into the model.

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \epsilon_i = \sum_{k=1}^p X_{ik} \beta_k + \epsilon_i$$

- Here  $X_{1i} = 1$  typically, so that an intercept is included.
- Least squares minimizes

$$\sum_{i=1}^n \left( Y_i - \sum_{k=1}^p X_{ki} \beta_k \right)^2$$

# How to get estimates

- With two regressors,  $E[Y_i] = X_{1i}\beta_1 + X_{2i}\beta_2 = \mu_i$ .
- Least squares tries to minimize

$$\sum_{i=1}^n (Y_i - X_{1i}\beta_1 - X_{2i}\beta_2)^2$$

# How does it work?

- Consider the model  $y = \alpha + \beta_1 x_1 + \beta_2 x_2 + e$ .
- We want to know the relationship between  $x_1$  and  $y$  adjusted for the effect of  $x_2$ .
  - 1 Fit a regression of  $y$  on  $x_2$  only and save the residuals — the part  $y$  not explained by  $x_2$ .
  - 2 Fit a regression of  $x_1$  on  $x_2$  — the part of  $x_1$  not explained by  $x_2$ .
  - 3 Plot residuals from step 1 on residuals from step 2 — this gives us  $x_1$ 's conditional effect on  $y$ .

# Result



$$\hat{\beta}_1 = \frac{\sum_{i=1}^n e_{i,Y|X_2} e_{i,X_1|X_2}}{\sum_{i=1}^n e_{i,X_1|X_2}^2}$$

- That is, the regression estimate for  $\beta_1$  is the regression through the origin estimate having regressed  $X_2$  out of both the response and the predictor.
- (Similarly, the regression estimate for  $\beta_2$  is the regression through the origin estimate having regressed  $X_1$  out of both the response and the predictor.)
- More generally, multivariate regression estimates have removed (or accounted for) the linear relationship between the other variables from both the regressor and response.

# The general case

- Least squares solutions have to minimize

$$\sum_{i=1}^n (Y_i - X_{1i}\beta_1 - X_{2i}\beta_2 - \dots - X_{pi}\beta_p)^2$$

- The least squares estimate for the coefficient of a multivariate regression model can be interpreted as the result of regressing the outcome on that predictor after removing the linear effects of the other predictors from both the outcome and the predictor.
- In this sense, multivariate regression “adjusts” a coefficient for the linear impact of the other variables.

# Interpretation of the coefficients

So that the interpretation of a multivariate regression coefficient is the expected change in the response per unit change in the regressor, holding all of the other regressors fixed.

# Dummy Variables

- What if we wish to include a dichotomous variable as a regressor in a multivariate OLS model?
- How do we interpret the coefficient?

# Dummy variables are smart

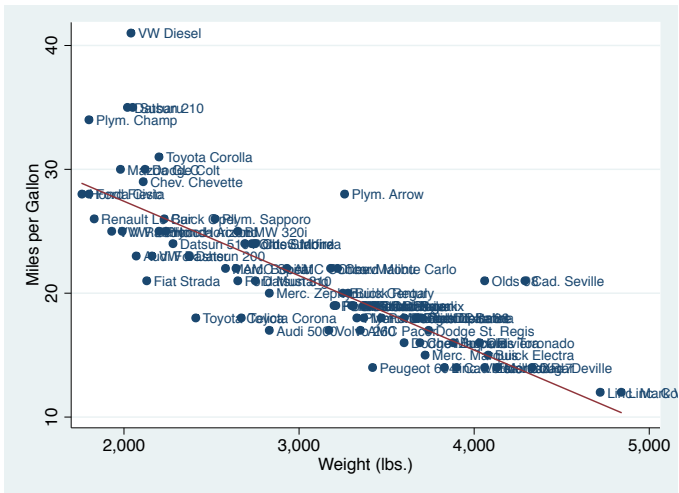
Consider the linear model

$$Y_i = \beta_0 + X_{i1}\beta_1 + \epsilon_i$$

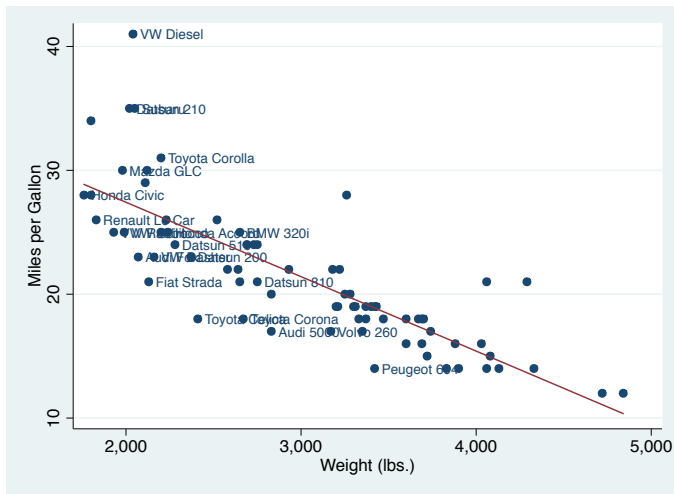
where each  $X_{i1}$  is binary so that it is a 1 if measurement  $i$  is in a group and 0 otherwise. (Treated versus not in a clinical trial, for example.)

- Then for people in the group  $E[Y_i] = \beta_0 + \beta_1$
- And for people not in the group  $E[Y_i] = \beta_0$
- The LS fits work out to be  $\hat{\beta}_0 + \hat{\beta}_1$  is the mean for those in the group and  $\hat{\beta}_0$  is the mean for those not in the group.
- $\beta_1$  is interpreted as the increase or decrease in the mean comparing those in the group to those not.
- Note including a binary variable that is 1 for those not in the group would be redundant. It would create three parameters to describe two means.

## Dummy Variables: MPG and Weight



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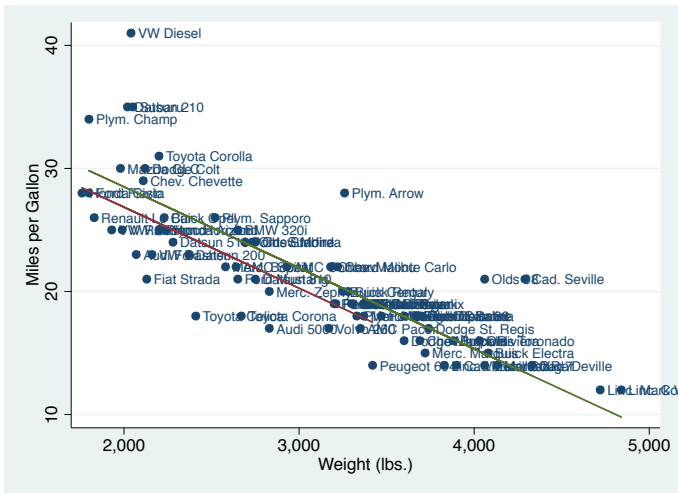
# MPG Regression with Foreign Dummy

|              | (1)                  |
|--------------|----------------------|
|              | mpg                  |
| Weight       | -0.007***<br>(0.001) |
| Foreign      | -1.650<br>(1.076)    |
| Constant     | 41.680***<br>(2.166) |
| Observations | 74                   |
| R-squared    | 0.663                |

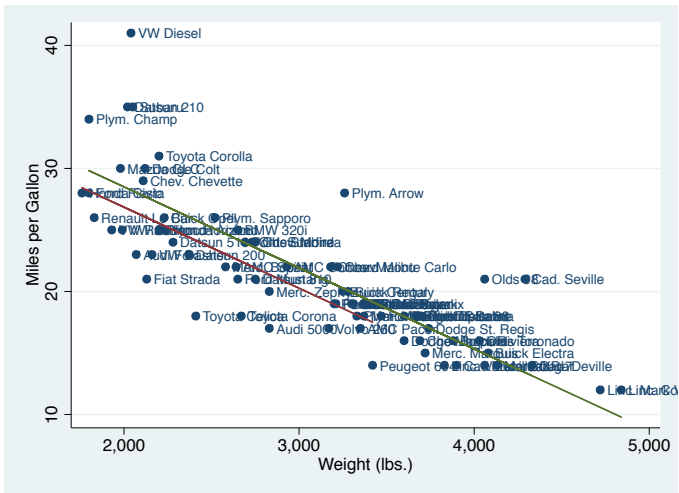
Standard errors in parentheses

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

# Dummy Variables: MPG and Weight



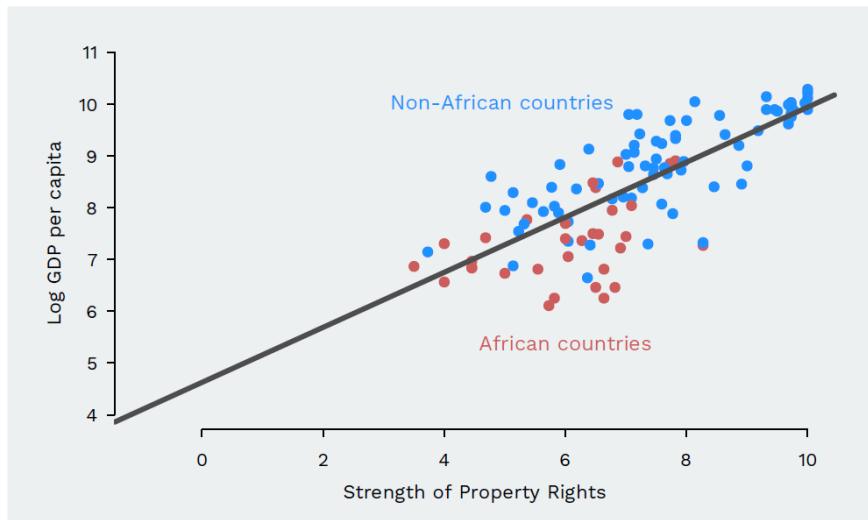
## Dummy Variables: MPG and Weight



## More than 2 levels

- Consider a multilevel factor level. Let's say a three level factor (example, US political party affiliation: Republican, Democrat, Independent)
- $Y_i = \beta_0 + X_{i1}\beta_1 + X_{i2}\beta_2 + \epsilon_i$ .
- $X_{i1}$  is 1 for Republicans and 0 otherwise.
- $X_{i2}$  is 1 for Democrats and 0 otherwise.
- If  $i$  is Republican  $E[Y_i] = \beta_0 + \beta_1$
- If  $i$  is Democrat  $E[Y_i] = \beta_0 + \beta_2$ .
- If  $i$  is Independent  $E[Y_i] = \beta_0$ .
- $\beta_1$  compares Republicans to Independents.
- $\beta_2$  compares Democrats to Independents.
- $\beta_1 - \beta_2$  compares Republicans to Democrats.
- (Choice of reference category changes the interpretation.)

# Risk Example



## Risk Example

- Suppose I'm interested in the relationship between property rights and development.
- However, African countries might have low incomes and weak property rights
- Conditioning  $Y$  on a country being in Africa or not to remove this

$$E[Y_i|X_i, Z_i] = \beta_0 + \beta_1 X_i + \beta_2 Z_i$$

- $Z = 1$ , if  $i$  is an African country, else  $Z = 0$ .
- Effects are now within Africa or within non-Africa, not between

## Risk Example

```
ajr.mod <- lm(logpgp95 ~ avexpr + africa, data = ajr)
summary(ajr.mod)
##
## Coefficients:
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) 5.6556 0.3134 18.04 <2e-16 ***
## avexpr 0.4242 0.0397 10.68 <2e-16 ***
## africa -0.8784 0.1471 -5.97 3e-08 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.625 on 108 degrees of freedom
## (52 observations deleted due to missingness)
## Multiple R-squared: 0.708, Adjusted R-squared: 0.702
## F-statistic: 131 on 2 and 108 DF, p-value: <2e-16
```

## Risk Example Interpretation

- How can we interpret this model?
- Plug in two possible values for  $Z$  and rearrange

When  $Z = 0$  (non-African countries)

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 Z_i \quad (1)$$

$$= \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 \times 0 \quad (2)$$

$$= \hat{\beta}_0 + \hat{\beta}_1 X_i \quad (3)$$

When  $Z = 1$  (African countries)

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 Z_i \quad (4)$$

$$= \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 \times 1 \quad (5)$$

$$= (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 X_i \quad (6)$$

Two different intercepts, same slope for each group

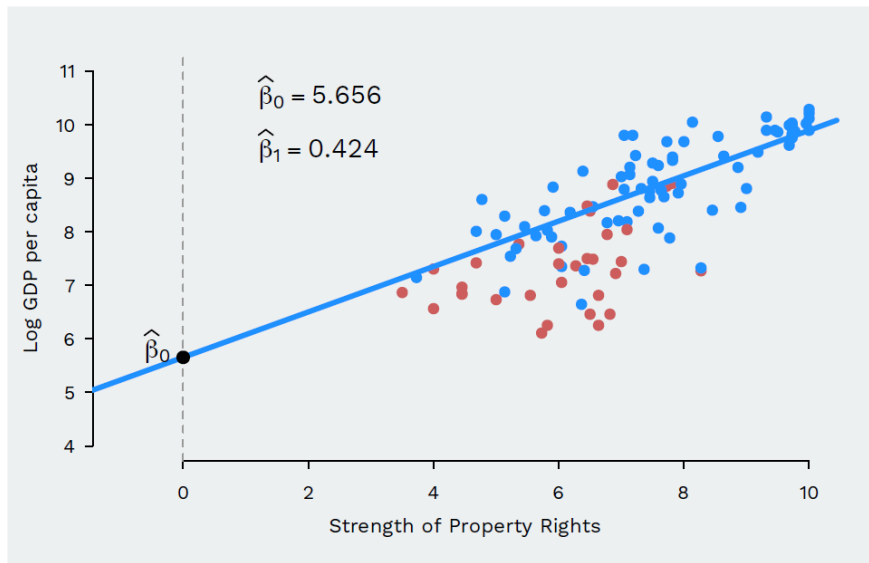
## Risk Example Interpretation

- In this example, we have:

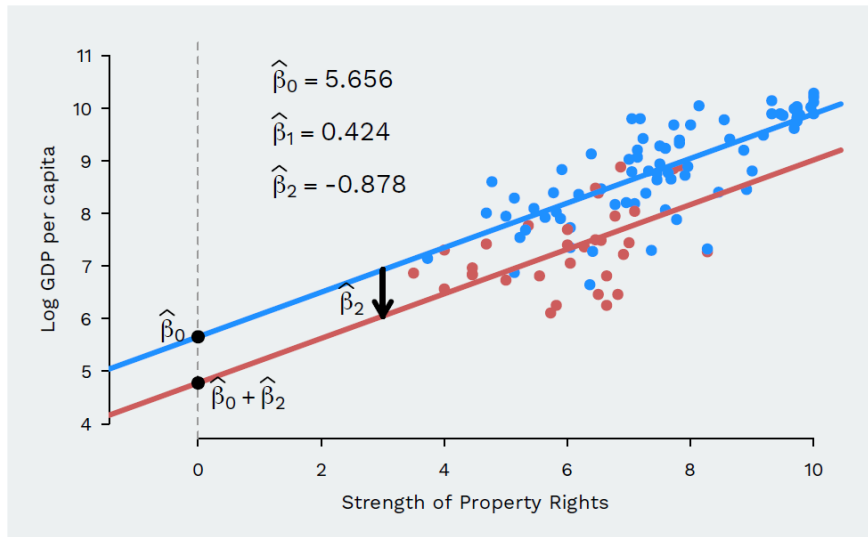
$$\hat{Y}_i = 5.656 + 0.424 * X_i - 0.878 * Z_i$$

- $\hat{\beta}_0$ : average log income for non-African country ( $Z = 0$ ) with property rights measured at 0 is 5.656.
- $\hat{\beta}_1$ : A one-unit increase in property rights is associated with a 0.424 increase in average log incomes for two African countries or for two non-African countries (generally, controlling for units with the same value of  $Z$ ).
- $\hat{\beta}_2$ : there is a -0.878 average difference in log income per capita between African and non-African counties conditional on property rights (generally, controlling for units with the same value of  $X$ ).

# Risk Example



# Risk Example



# Adding a continuous variable

- We're still interested in the relationship between property rights and development.
- Why would Africa be a potential confounder?
  - ▶ Geography/climate might affect political institutions.
  - ▶ Geography/climate might affect average incomes (through reduced productivity because of diseases like malaria)
- Condition on Z: mean temperature in country  $i$  (continuous variable)

## Risk Example

```
ajr.mod2 <- lm(logpgp95 ~ avexpr + meantemp, data = ajr)
summary(ajr.mod2)
##
## Coefficients:
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) 6.8063 0.7518 9.05 1.3e-12 ***
## avexpr 0.4057 0.0640 6.34 3.9e-08 ***
## meantemp -0.0602 0.0194 -3.11 0.003 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.643 on 57 degrees of freedom
## (103 observations deleted due to missingness)
## Multiple R-squared: 0.615, Adjusted R-squared: 0.602
## F-statistic: 45.6 on 2 and 57 DF, p-value: 1.48e-12
```

## Risk Example Interpretation

$$\hat{Y}_i = \hat{\beta}_{0i} + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i}$$

$$\hat{Y}_i = 7.527 + 0.327 \times \text{PR Risk} - 0.689 \times \text{Africa} - 0.056 \times \text{Avg. Temp}$$

- $\hat{\beta}_0$ : average log income for non-African country ( $Z = 0$ ) with property rights measured at 0 and mean temp is zero is 7.527 (not meaningful).
- $\hat{\beta}_1$ : A one-unit increase in property rights is associated with a 0.327 increase in average log incomes conditional on a mean temperature and geographic location (generally, controlling for units with the same value of  $X_2$  and  $X_3$ ).
- $\hat{\beta}_2$ : there is a -0.689 average difference in log income per capita between African and non-African counties conditional on property rights and temperature (generally, controlling for units with the same value of  $X_1$  and  $X_3$ ).
- $\hat{\beta}_3$ : A one-degree increase in mean temperature is associated with a 0.056 change in average log incomes conditional on strength of property rights (generally, controlling for units with the same value of  $X_1$  and  $X_2$ ).

# **APPENDIX**

## Interpretation of the coefficients

$$E[Y|X_1 = x_1, \dots, X_p = x_p] = \sum_{k=1}^p x_k \beta_k$$

$$E[Y|X_1 = x_1 + 1, \dots, X_p = x_p] = (x_1 + 1)\beta_1 + \sum_{k=2}^p x_k \beta_k$$

(k=2 because I've pulled k=1 to the front of the summation)

$$E[Y|X_1 = x_1 + 1, \dots, X_p = x_p] - E[Y|X_1 = x_1, \dots, X_p = x_p]$$

$$= (x_1 + 1)\beta_1 + \sum_{k=2}^p x_k \beta_k - \sum_{k=1}^p x_k \beta_k$$

$$= (x_1 + 1)\beta_1 + \sum_{k=2}^p x_k \beta_k - ((x_1)\beta_1 + \sum_{k=2}^p x_k \beta_k)$$

$$(x_1 + 1)\beta_1 - (x_1)\beta_1 = \beta_1$$